Building Trust in Scientific Data: CoreTrustSeal Certification & the World Data System

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Why Build Trust in Scientific Data?

- There are many technical barriers to data sharing
- But…being unable to trust data from other sources is one of the major challenges preventing proper data management, preservation, and sharing

Trust is thus at the very heart of storing and sharing data:

- Data funders want reassurances their investment is not wasted; whilst data reuse will give them a higher return on investment
- Data depositors want to be sure their data are safe and remain accessible, usable, and meaningful over time
- Data users want to know that data have been preserved properly and are of high quality

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How to Build Trust in Scientific Data?

- Certification standards play an important role in establishing trust, and hence sustaining the opportunities for long-term data sharing
- The demand to evaluate the trustworthiness of data repositories has been answered by three levels of certification services becoming available
  - **Core Certification:** CoreTrustSeal (16 criteria)
  - **Extended Certification:** DIN 31644 / nestor Seal (34 criteria)
  - **Formal Certification:** ISO 16363 (100+ criteria & an external audit)
- These standards take into account technical infrastructure and standards; organizational, financial, staffing, and legal aspects; as well as workflows, risk management, etc.

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What is CoreTrustSeal Certification?

- CoreTrustSeal certification is a minimally intensive process whereby a data repository provides evidence it is sustainable and trustworthy via an online application of 16 Requirements
- Evidence is given by supplying URLs of (ideally public) documents
- Applicants also indicate a Compliance Level for each Requirement to assist their own understanding of their maturity
- Payment of a 1000 EUR administration fee allows an applications to be reviewed by two community peers
- Successful applications are made publicly available
- Renewal is every three years
- Note: Completing a self-assessment is very useful even without certification as it enables appraisal of internal procedures

What are the 16 Requirements?

- **Background information**
  - Context
- **Organizational infrastructure**
  - Mission/scope
  - Licenses
  - Continuity of access
  - Confidentiality and ethics
  - Organizational infrastructure
  - Expert guidance

- **Digital object management**
  - Data integrity and authenticity
  - Appraisal
  - Documented storage procedures
  - Preservation plan
  - Data quality
  - Workflows
  - Data discovery and identification
  - Data reuse
- **Technology**
  - Technical infrastructure
  - Security

How Does a Requirement Look?

XIV. Data reuse

R14. The repository enables reuse of the data over time, ensuring that appropriate metadata are available to support the understanding and use of the data.

Compliance Level ☆☆☆☆

Response

Guidance:
Repositories must ensure that data can be understood and used effectively into the future despite changes in technology. This Requirement evaluates the measures taken to ensure that data are reusable.

For this Requirement, responses should include evidence related to the following questions:

- Which metadata are required by the repository when the data are provided (e.g., Dublin Core or content-oriented metadata)?
- Are data provided in formats used by the Designated Community? Which formats?
- Are measures taken to account for the possible evolution of formats?
- Are plans related to future migrations in place?
- How does the repository ensure understandability of the data?

Reuse is dependent on the applicable licenses covered in R2 (Licenses).

Note: Certification can be granted even if some Requirements are not at Level 4

A brief history of ADA

The Social Science Data Archive (now ADA) was set up in 1981, housed in the Research School of Social Sciences at the Australian National University

Mission: provide a national service for the collection and preservation of digital research data and to make these data available for secondary analysis by academic researchers and other users.

The Archive now sits in Centre for Social Research and Methods at The ANU.

The Archive holds over 5000 datasets from around 1500 studies, including national election studies; public opinion polls; social attitudes surveys, censuses, aggregate statistics, administrative data and many other sources.

Data holdings are sourced from academic, government and private sectors.

Serviced by National Computational Infrastructure.

DSA/WDS Process

Received funding from ANDS to undertake a study to assess ADA's position as a "trusted repositories"

- The Australian National Data Service (ANDS) makes Australia's research data assets more valuable for researchers, research institutions and the nation.

Initially looked at documentation using 2014-2017 ("pre-WDS") assessment guidelines

Following publication in September 2016, decided to focus on requirements for new (DSA/WDS) standard

Started the "official" DSA/WDS process in March

- Submitted early April
- Review due late May arrived end August

Used new (November 2016) criteria of DSA/WDS

New name/website same criteria Core Trust Seal

Core Trust Seal Outcomes

Identified 4 guidelines set at level 3

- 7 Data integrity and authenticity
- 10 Preservation plan
- 15 Technical infrastructure
- 16 Security

Identified 12 guidelines set at level 4

- Assessor indicated potential to move Guideline 9 Documented storage procedures to a 3

Challenges in the Process

What are the “minimum” requirements?

- Previously have recommendation on minimum
- Current no advice on appropriate or expected level – our assumption is that you need at least a “1”
- Do you have to respond to all the guidelines?

Complex interplay between guidelines and relevant documents

How to provide evidence from documents/items not in the public domain

Assessor wants timelines for “in process of implementation”

9 of 16 guidelines required edits. In addition most links required website updates due to changes.

Process timeframes and outcomes have been adressed

Many of these challenges have been rectified by the feedback loop inherent in the process.

Australian context

Research Data Australia (ANDS discovery service) lists 103 “Research organisations”

Mostly

- Institutional repositories
- Disciplinary repositories/archives
- Government agencies

Australian context

Considerations for the application of DSA/WDS:

Complexity of Institutions and Repositories

National Frameworks

Funding Frameworks

Infrastructure Frameworks

Governance frameworks

Our assessment: These are not Australian-specific challenges, but reflections of the diversity of the “repository sector”.

Question: does Core Trust Seal adequately support the variation in repository types that exist within the above frameworks?

Recommendations from our experience

Who is your community and do they require TRUST/FAIR

Detailing items in public domain

How to cite items out of public domain

- Should we move “private” elements into the public domain?

How to explain items out of our direct control

- E.g. funding, infrastructure, governance

Risk management standards

- Where to access?

Changes Implemented at ADA

ADA AUSTRALIAN DATA ARCHIVE

Website

Documentation

Service Platform for repository

Self Deposit and ensuring FAIR and TRUST and 5 Safes

Collaborations

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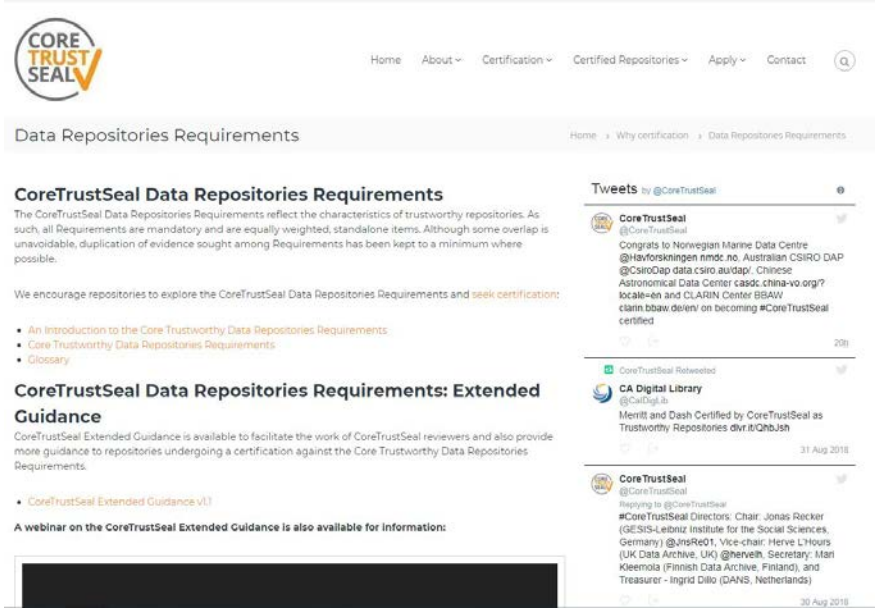
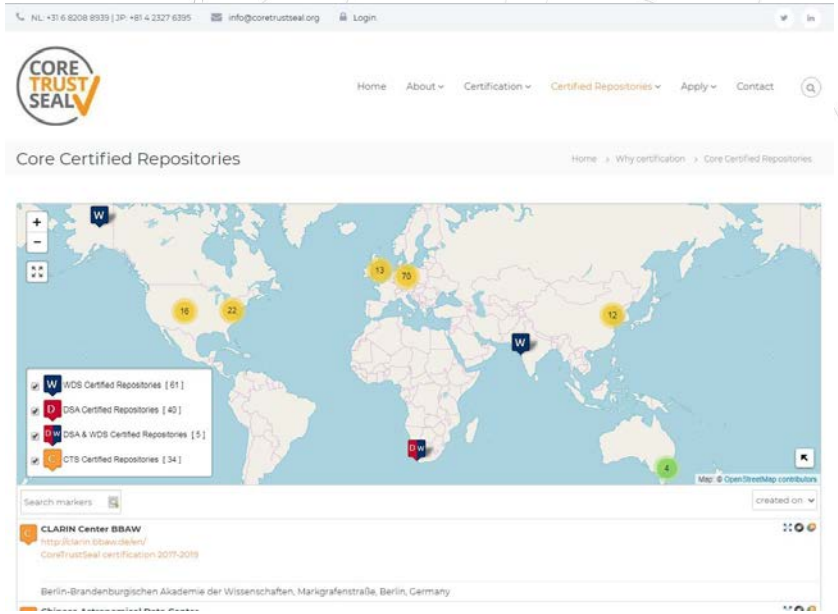
Are There Resources to Help Me?





www.coretrustseal.org/why-certification/requirements/

- Extended Guidance and webinars.
- The Extended Guidance is intended for reviewers, but is useful for applicants.

https://www.coretrustseal.org/why-certification/certified-repositories/

Library of public applications. All are certified and so can be considered exemplars.

What is the World Data System?

- The World Data System (WDS) is an Interdisciplinary Body of the International Science Council (ISC)
- WDS promotes **long-term stewardship** of, and **universal and equitable access** to, **quality-assured** scientific data, data services, products, and information across <u>all disciplines and geographies</u>
- To fulfil its remit, WDS is building worldwide '**communities of excellence**' of Trustworthy Data Services by accrediting Member Organizations

76 Regular	Data stewards and/or data analysis services: Trustworthy Data Repositories	Certified
11 Networks	Umbrella bodies of data stewards: Trustworthy Data Networks	Certified
11 Partners	Contribute support to WDS membership	Co-opted
19 Associates	Interested in the WDS endeavour	Co-opted

Why Join WDS?

- **IT IMPROVES YOUR LOCAL AND INTERNATIONAL RECOGNITION**
  - Heightened profile
  - Increased visibility and engagement
  - Improved funding prospects
  - Facilitated interactions and data exchange with the international community
- **IT DEMONSTRATES YOUR COMMITMENT TO OPEN SCIENCE**
  - Adherence to the WDS Data Sharing Principles
  - Promoting your organization as a trustworthy scientific data service
- **IT INCREASES YOUR PERFORMANCE AND AGILITY**
  - Enhanced discovery of data and increased (inter)disciplinary and user base
  - Improved practices and processes
  - Facilitated citation of data

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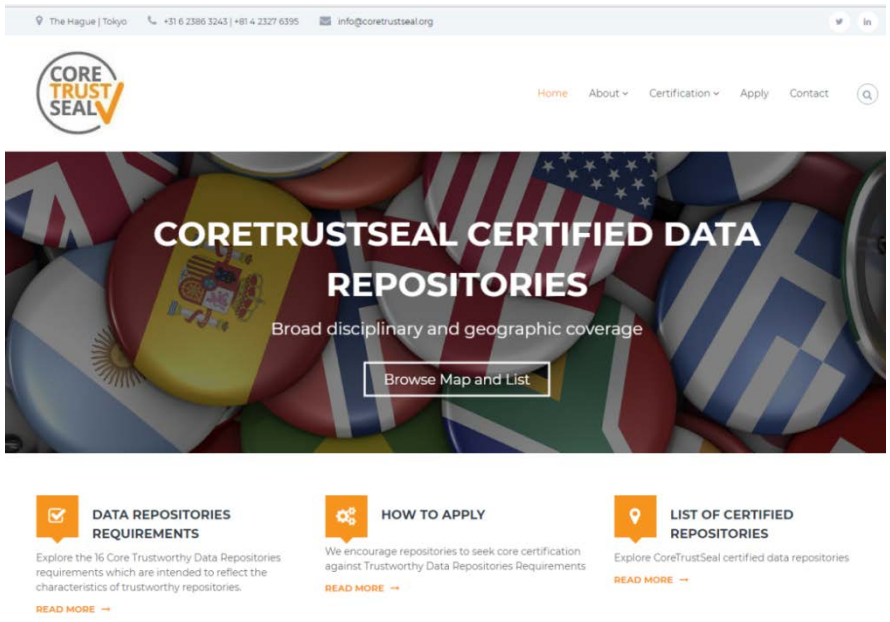
Where Can I Find More Information?

www.CoreTrustSeal.org



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